

강의노트 109쪽의 8.21 식으로부터 시작한다.

$$U = \frac{\mu}{r} \left[1 - \sum_{j=2}^{\infty} J_4 \left(\frac{R_0}{r} \right)^4 P_4 [\sin \vartheta_{gsat}] \right] + \sum_{j=2}^{\infty} \sum_{m=1}^j \left(\frac{R_0}{r} \right)^j P_{0,m} [\sin \vartheta_{gsat}] \left\{ C_{j,m} \cos(m\lambda_{sat}) + S_{j,m} \sin(m\lambda_{sat}) \right\}$$

위의 식에서 Zonal Harmonics ($m=0$)의 $J_2, J_3, \dots, J_6$ 를 2차원 disturbing 함수를 쓰면

$$\begin{aligned} R_{J_2, \dots, J_6} &= U_{J_2, \dots, J_6} - U_{2-body} = \left[-\frac{\mu}{r} \right] \left[\sum_{j=2}^6 J_4 \left(\frac{R_0}{r} \right)^4 P_4 [\sin \vartheta_{gsat}] \right] \\ &= -\frac{\mu}{r} \left(J_2 \left(\frac{R_0}{r} \right)^2 P_2 [\sin \vartheta_{gsat}] + J_3 \left(\frac{R_0}{r} \right)^3 P_3 [\sin \vartheta_{gsat}] + \dots + J_6 \left(\frac{R_0}{r} \right)^6 P_6 [\sin \vartheta_{gsat}] \right) \\ &= -\frac{\mu J_2}{r} \left(\frac{R_0}{r} \right)^2 \frac{3}{2} \left(\sin^2 \vartheta - \frac{1}{3} \right) - \frac{\mu J_3}{r} \left(\frac{R_0}{r} \right)^3 \frac{5}{2} \left(\sin^3 \vartheta - \frac{3}{5} \sin \vartheta \right) - \frac{\mu J_4}{r} \left(\frac{R_0}{r} \right)^4 \frac{35}{8} \left(\sin^4 \vartheta - \frac{20}{35} \sin^2 \vartheta + \frac{3}{35} \right) \\ &\quad - \frac{\mu J_5}{r} \left(\frac{R_0}{r} \right)^5 \frac{63}{8} \left(\sin^5 \vartheta - \frac{70}{63} \sin^3 \vartheta + \frac{65}{63} \sin \vartheta \right) - \frac{\mu J_6}{r} \left(\frac{R_0}{r} \right)^6 \frac{231}{16} \left(\sin^6 \vartheta - \frac{315}{231} \sin^4 \vartheta + \frac{105}{231} \sin^2 \vartheta - \frac{5}{231} \right) \end{aligned}$$

위의 식에 아래의 구면삼각형에서의 사인법칙을 이용한다.



$$\Rightarrow \frac{\sin(\omega+\nu)}{\sin 90} = \frac{\sin \vartheta}{\sin \lambda}$$

$$\text{그러나 } \sin \vartheta = \sin j \cdot \sin(\omega+\nu) \text{ 임을 이용}$$

따라서 정리한 식은 다음과 같다.

$$\begin{aligned} & -\frac{\mu J_2}{r} \left(\frac{R_0}{r} \right)^2 \frac{1}{2} (3 \sin^2 \lambda \sin^2(\omega+\nu) - 1) - \frac{\mu J_3}{r} \left(\frac{R_0}{r} \right)^3 \frac{1}{2} (5 \sin^3 \lambda \sin^3(\omega+\nu) - 3 \sin \lambda \sin(\omega+\nu)) \\ & - \frac{\mu J_4}{r} \left(\frac{R_0}{r} \right)^4 \frac{1}{8} (35 \sin^4 \lambda \sin^4(\omega+\nu) - 30 \sin^2 \lambda \sin^2(\omega+\nu) + 2) \\ & - \frac{\mu J_5}{r} \left(\frac{R_0}{r} \right)^5 \frac{1}{8} (63 \sin^5 \lambda \sin^5(\omega+\nu) - 70 \sin^3 \lambda \sin^3(\omega+\nu) + 15 \sin \lambda \sin(\omega+\nu)) \\ & - \frac{\mu J_6}{r} \left(\frac{R_0}{r} \right)^6 \frac{1}{16} (231 \sin^6 \lambda \sin^6(\omega+\nu) - 315 \sin^4 \lambda \sin^4(\omega+\nu) + 105 \sin^2 \lambda \sin^2(\omega+\nu) - 5) \end{aligned}$$

삼각함수의 반각 (배각) 공식을 참고하여 식을 전개한다.

$$\sin^2(\omega + \nu) = \frac{1 - \cos 2(\omega + \nu)}{2} \quad \dots \textcircled{1}$$

$$\sin^3(\omega + \nu) = \frac{3\sin(\omega + \nu) - \sin 3(\omega + \nu)}{4} \quad \dots \textcircled{2}$$

$$\sin^4(\omega + \nu) = \textcircled{1}^2 = \frac{1 - 2\cos 2(\omega + \nu) + \cos^2 2(\omega + \nu)}{4} = \frac{3 - 4\cos 2(\omega + \nu) + \cos 4(\omega + \nu)}{8}$$

$$\sin^5(\omega + \nu) = \textcircled{1} \cdot \textcircled{2} = \frac{3\sin(\omega + \nu) - 3\sin 3(\omega + \nu) - \frac{3}{2}(\sin 3(\omega + \nu) - \sin(\omega + \nu)) + \frac{1}{2}(\sin 5(\omega + \nu) - \sin(\omega + \nu))}{8}$$

$$= \frac{10\sin(\omega + \nu) - 9\sin 3(\omega + \nu) + \sin 5(\omega + \nu)}{16}$$

$$\sin^6(\omega + \nu) = \textcircled{2}^2 = \frac{9\sin^2(\omega + \nu) - 6\sin(\omega + \nu)\sin 3(\omega + \nu) + \sin^2 3(\omega + \nu)}{16}$$

$$= \frac{9(1 - \cos 2(\omega + \nu)) + 6(\cos 4(\omega + \nu) - \cos 2(\omega + \nu)) + (1 - \cos 6(\omega + \nu))}{32}$$

이 표 제이지의 식에 이를 적용하면

$$\Rightarrow -\frac{3\mu J_2}{2} \left(\frac{R^2}{a^3} \right) \left(\frac{a}{r} \right)^3 \left(\frac{1}{2} \sin^2 i - \frac{1}{2} \sin^2 i \cos 2(\omega + \nu) - \frac{1}{3} \right)$$

$$- \frac{\mu J_2}{2} \left(\frac{R^3}{a^3} \right) \left(\frac{a}{r} \right)^4 \left(\frac{15}{4} \sin^3 i \sin(\omega + \nu) - \frac{5}{4} \sin^3 i \sin 3(\omega + \nu) - 3i \sin(\omega + \nu) \right)$$

$$- \frac{\mu J_4}{8} \left(\frac{R^4}{a^5} \right) \left(\frac{a}{r} \right)^5 \left(\frac{105}{32} \sin^4 i - \frac{35}{8} \sin^4 i \cos 2(\omega + \nu) + \frac{36}{32} \sin^4 i \cos 4(\omega + \nu) - 15 \sin^2 i + \right.$$

$$\left. - \frac{\mu J_6}{8} \left(\frac{R^5}{a^5} \right) \left(\frac{a}{r} \right)^6 \left(\frac{63.5}{8} \sin^5 i \sin(\omega + \nu) - \frac{63.9}{16} \sin^5 i \sin 3(\omega + \nu) + \frac{15}{16} \sin^2 i \cos 2(\omega + \nu) + 3 \right) \right.$$

$$\left. - \frac{63.35}{42} \sin^3 i \sin(\omega + \nu) + \frac{63.35}{126} \sin^3 i \sin 3(\omega + \nu) + 15 \sin i \sin(\omega + \nu) \right)$$

$$- \frac{\mu J_6}{16} \left(\frac{R^6}{a^6} \right) \left(\frac{a}{r} \right)^7 \left(\frac{1155}{16} \sin^6 i - \frac{3465}{32} \sin^6 i \cos 2(\omega + \nu) + \frac{693}{16} \sin^6 i \cos 4(\omega + \nu) - \frac{231}{32} \sin^6 i \cos 6(\omega + \nu) \right)$$

$$- \frac{945}{32} \sin^4 i + \frac{315}{8} \sin^4 i \cos 2(\omega + \nu) - \frac{315}{32} \sin^4 i \cos 4(\omega + \nu)$$

$$+ \frac{105}{2} \sin^2 i - \frac{105}{2} \sin i \cos 2(\omega + \nu) - 5 \Big)$$

secular term은 ω 와 ν 를 무시한 항이므로 다시 정리하면

$$\begin{aligned}
 R_{sec J_2 \dots J_6} = & -\frac{3\pi J_2}{2} \left(\frac{R^2}{a^3}\right) \left(\frac{a}{r}\right)^3 \left(\frac{1}{2} \sin^2 i - \frac{1}{3}\right) \\
 & -\frac{35\pi J_4}{8} \left(\frac{R^4}{a^5}\right) \left(\frac{a}{r}\right)^5 \left(\frac{3}{32} \sin^4 i - \frac{3}{11} \sin^2 i + \frac{3}{35}\right) \\
 & -\frac{231\pi J_6}{16} \left(\frac{R^6}{a^7}\right) \left(\frac{a}{r}\right)^7 \left(\frac{5}{16} \sin^6 i - \frac{45}{352} \sin^4 i + \frac{5}{22} \sin^2 i - \frac{5}{231}\right)
 \end{aligned}$$

이번엔 강타노드 110쪽을 참고하면 $\left(\frac{a}{r}\right)$ 이 관한 식을 얻을 수 있다.

$$\Rightarrow M = \bar{E} - e \sin E, \quad dM = (1 - e \cos E) dE \Rightarrow \frac{dE}{dM} = \frac{1}{1 - e \cos E} = \frac{a}{r}$$

$$r = \frac{a(1-e^2)}{1 + e \cos \nu} = a(1 - e \cos E)$$

$$\frac{d}{d\nu} \left(\frac{a(1-e^2)}{1 + e \cos \nu} \right) = \frac{d}{dE} (a(1 - e \cos E))$$

$$\frac{a(1-e^2)}{1 + e \cos \nu} \cdot \frac{e \sin \nu}{1 + e \cos \nu} d\nu = a e \sin E dE \quad \leftarrow \sin E = \frac{\sqrt{1-e^2} \sin \nu}{1 + e \cos \nu}$$

$$\therefore \frac{d\nu}{dE} = \frac{a}{r} \sqrt{1-e^2}, \quad \frac{d\nu}{dM} = \frac{d\nu}{dE} \cdot \frac{dE}{dM} = \left(\frac{a}{r}\right)^2 \sqrt{1-e^2}$$

$$\overline{\left(\frac{a}{r}\right)^2} = \frac{\int_0^{2\pi} \left(\frac{a}{r}\right)^2 dM}{\int_0^{2\pi} dM} = \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{a}{r}\right)^3 dM = \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{a}{r}\right)^3 \frac{dM}{d\nu} d\nu = \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{a}{r}\right)^3 \left(\frac{r}{a}\right)^2 \frac{1}{\sqrt{1-e^2}} d\nu$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \frac{a}{r} \frac{1}{\sqrt{1-e^2}} d\nu = \frac{1}{2\pi} \int_0^{2\pi} \frac{1 + e \cos \nu}{1 - e^2} \frac{1}{\sqrt{1-e^2}} d\nu = \frac{1}{2\pi(1-e^2)^{\frac{3}{2}}} \int_0^{2\pi} (1 + e \cos \nu) d\nu = \frac{1}{(1-e^2)^{\frac{3}{2}}}$$

$$\overline{\left(\frac{a}{r}\right)^5} = \frac{\int_0^{2\pi} \left(\frac{a}{r}\right)^5 dM}{\int_0^{2\pi} dM} = \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{a}{r}\right)^5 dM = \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{a}{r}\right)^5 \frac{dM}{d\nu} d\nu = \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{a}{r}\right)^5 \left(\frac{r}{a}\right)^2 \frac{1}{\sqrt{1-e^2}} d\nu$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{1 + e \cos \nu}{1 - e^2}\right)^3 \frac{1}{\sqrt{1-e^2}} d\nu = \frac{1}{2\pi(1-e^2)^{\frac{7}{2}}} \int_0^{2\pi} (1 + e \cos \nu)^3 d\nu = \frac{1 + \frac{3}{2}e^2}{(1-e^2)^{\frac{7}{2}}}$$

$$\overline{\left(\frac{a}{r}\right)^7} = \frac{\int_0^{2\pi} \left(\frac{a}{r}\right)^7 dM}{\int_0^{2\pi} dM} = \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{a}{r}\right)^7 dM = \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{a}{r}\right)^7 \frac{dM}{d\nu} d\nu = \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{a}{r}\right)^7 \left(\frac{r}{a}\right)^2 \frac{1}{\sqrt{1-e^2}} d\nu$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{1 + e \cos \nu}{1 - e^2}\right)^5 \frac{1}{\sqrt{1-e^2}} d\nu = \frac{1}{2\pi(1-e^2)^{\frac{9}{2}}} \int_0^{2\pi} (1 + e \cos \nu)^5 d\nu = \frac{1 + 5e^2 + 1.875e^4}{(1-e^2)^{\frac{9}{2}}}$$

이렇게 정리된 secular term을 다시 쓰면.

$$R_{sec, J_2, \dots, J_6} = -\frac{3MJ_2}{2} \left(\frac{R^4}{a^3} \right) \frac{1}{(1-e^2)^{\frac{3}{2}}} \left(\frac{1}{2} \sin^2 i - \frac{1}{3} \right) \\ - \frac{MJ_4}{8} \left(\frac{R^4}{a^5} \right) \frac{1 + \frac{3}{2}e^2}{(1-e^2)^{\frac{5}{2}}} \left(\frac{105}{32} \sin^4 i - 15 \sin^2 i + 3 \right) \\ - \frac{MJ_6}{16} \left(\frac{R^6}{a^7} \right) \frac{1 + 5e^2 + 1.875e^4}{(1-e^2)^{\frac{7}{2}}} \left(\frac{1155}{16} \sin^6 i - \frac{945}{32} \sin^4 i + \frac{105}{2} \sin^2 i - 5 \right)$$

위의 식을 Lagrange Planetary 4에 대입하면

$$\frac{d\Omega}{dt} = \frac{1}{na^2 \sqrt{1-e^2} \sin i} \frac{\partial R}{\partial i} \Rightarrow \frac{\partial R_{avg}}{\partial i} = -\frac{3MJ_2}{2} \frac{\left(\frac{R^2}{a^3} \right)}{(1-e^2)^{\frac{3}{2}}} \cdot 5 \sin i \cdot \cos i \\ - \frac{MJ_4}{8} \left(\frac{R^4}{a^5} \right) \frac{1 + \frac{3}{2}e^2}{(1-e^2)^{\frac{5}{2}}} \left(\frac{105}{8} \sin^4 i \cos i - 30 \sin^2 i \cos i \right) \\ - \frac{MJ_6}{16} \left(\frac{R^6}{a^7} \right) \frac{1 + 5e^2 + 1.875e^4}{(1-e^2)^{\frac{7}{2}}} \left(\frac{3465}{8} \sin^6 i \cos i - \frac{945}{8} \sin^4 i \cos i + 105 \sin^2 i \cos i \right)$$

이 때, $\mu = n^2 a^3$, $p = a(1-e^2)$ 을 대입하면

$$\frac{d\Omega}{dt} = -\frac{n}{p^2} \frac{3J_2 R_0^2 \cos i}{2} + \frac{n}{p^4} \frac{15J_4 R_0^4 \cos i}{32} \left(\frac{9}{8} - 12e^2 - (14 + 21e^2) \sin i \right) \\ - \frac{n}{p^6} \frac{105J_6 R_0^6 \cos i}{1024} \left(64 + 160e^2 + 120e^4 - (288 + 720e^2 + 540e^4) \sin i \right. \\ \left. + (264 + 660e^2 + 495e^4) \sin^2 i \right) \dots (4.9.38)$$

다항식이고 $\frac{d\omega}{dt} = \frac{\sqrt{1-e^2}}{na^2 e} \frac{\partial R}{\partial e} - \frac{\cos i}{na^2 \sqrt{1-e^2} \sin i} \frac{\partial R}{\partial i}$ 로 정리하면 공명이 발생한다.

$$\frac{d\omega}{dt} = -\frac{n}{p^2} \frac{3J_2 R_0^2}{4} (4 - 5 \sin^2 i) + \frac{n}{p^4} \frac{9J_4 R_0^4}{384} (56e^2 + (760 - 36e^2) \sin^2 i - (890 + 45e^2) \sin^4 i) \\ - \frac{n}{p^6} \frac{15J_6 R_0^6}{128} (64 + 72e^2 - (248 + 252e^2) \sin^2 i + (196 + 189e^2) \sin^4 i) \\ + \frac{n}{p^6} \frac{105J_6 R_0^6}{2048} (256 + 960e^2 + 320e^4 - (2048 + 6880e^2 + 2160e^4) \sin^2 i + (4128 + 13080e^2 + 3960e^4) \sin^4 i \\ - (2376 + 14520e^2 + 2145e^4) \sin^6 i) \dots (4.9.40)$$